

Lesson 01: Introduction to Differential Equations

Professor Dave Explains Differential Equations Master Series · Lesson 1

Video: Introduction to Differential Equations Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave introduces differential equations by contrasting them with ordinary algebra and standard calculus. In calculus, you are handed a curve and asked for its slope or curvature. In differential equations, nature hands you a rate equation—such as how speed changes with drag, or how cooling scales with temperature—and asks you to deduce the original motion or temperature curve. A differential equation defines an entire infinite family of curves filling the plane; it takes an initial condition (a single physical coordinate) to pin down the exact trajectory.

2. Core Theoretical Framework & Rigorous Formulations

Fundamental Mathematical Framework

An **ordinary differential equation (ODE)** expresses a relationship between an independent variable x , an unknown function $y(x)$, and its derivatives:

$$F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}\right) = 0$$

- Explicit Solution:** A function $\phi(x)$ defined on an interval I which, when substituted into the ODE, reduces it to an identity for all $x \in I$.
- General Solution:** An n -parameter family of functions $y = \phi(x, C_1, C_2, \dots, C_n)$ containing all possible particular solutions resulting from integration.
- Initial Value Problem (IVP):** An ODE paired with auxiliary constraints specified at a single point x_0 :

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1}$$

3. Fully Solved Whiteboard Problems

Classroom Problem: Deriving and Verifying the Trajectory of a Growth-Decay System

Problem Statement: Consider the initial value problem $\frac{dy}{dt} = -2y + 4$, with $y(0) = 5$. Determine the general solution family, verify that it identically satisfies the differential equation, and evaluate the specific particular solution satisfying the initial condition.

Step 1: Algebraic Separation of Variables

Rewrite the differential equation: $\frac{dy}{dt} = -2(y - 2) \implies \frac{1}{y-2} dy = -2dt$.

Step 2: Indefinite Integration Both Sides

$\int \frac{1}{y-2} dy = \int -2dt \implies \ln|y - 2| = -2t + C_1$.

Step 3: Exponentiation and Arbitrary Constant Absorption

$|y - 2| = e^{C_1} e^{-2t} \implies y(t) - 2 = A e^{-2t} \implies y(t) = 2 + A e^{-2t}$, where $A = \pm e^{C_1} \in \mathbb{R}$.

Step 4: Strict Identity Verification

Differentiate: $y'(t) = -2A e^{-2t}$. Evaluate RHS: $-2y + 4 = -2(2 + A e^{-2t}) + 4 = -4 - 2A e^{-2t} + 4 = -2A e^{-2t}$. Both sides match identically for all t .

Step 5: Enforcement of Initial Condition

$y(0) = 5 \implies 2 + A e^0 = 5 \implies A = 3$. Particular solution: $y(t) = 2 + 3e^{-2t}$.

⚠ Common Exam Pitfall & Trapdoor Avoidance

Never forget that the integration constant C_1 is exponentiated into $A = e^{C_1}$. Adding C outside the exponential (e.g. $y = 2 + e^{-2t} + C$) is a fatal algebraic error that fails the ODE.

4. Key Takeaways & Exam Strategy Formula Card

🎯 Core Takeaway & Exam Rule

ODEs generate families of continuous trajectories. The initial condition chooses the unique physical path through state space.

- Video reference: [Introduction to Differential Equations](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.